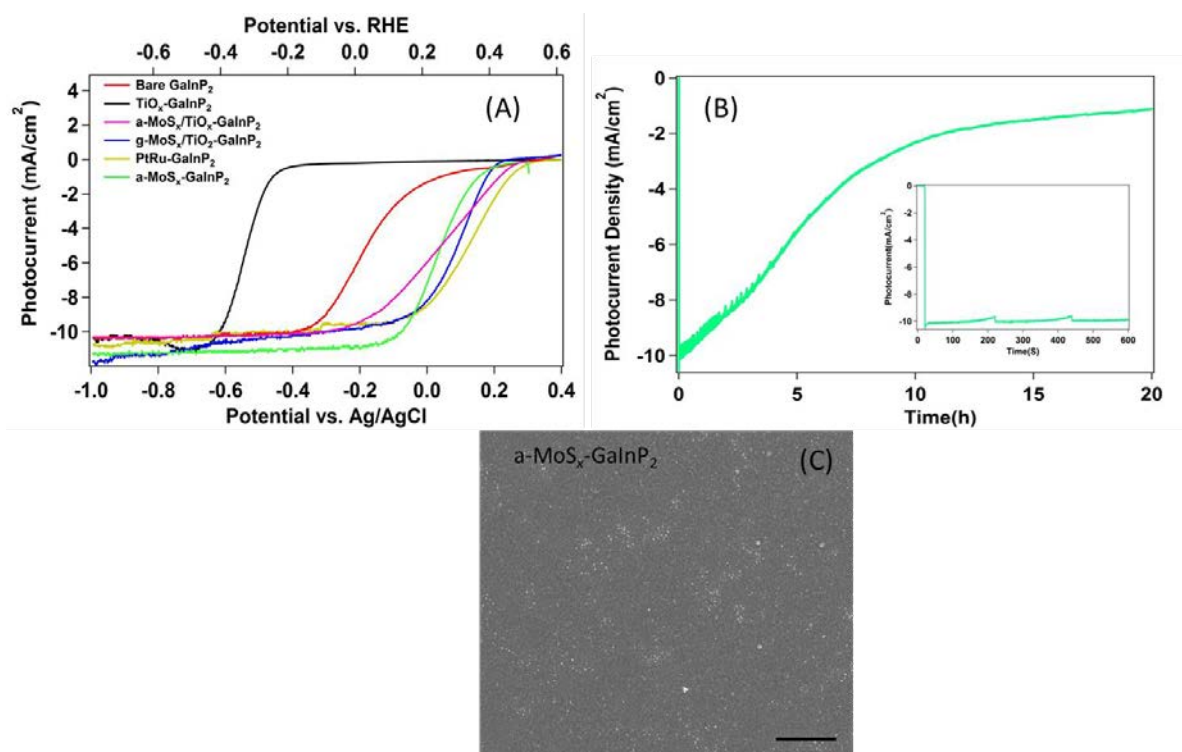


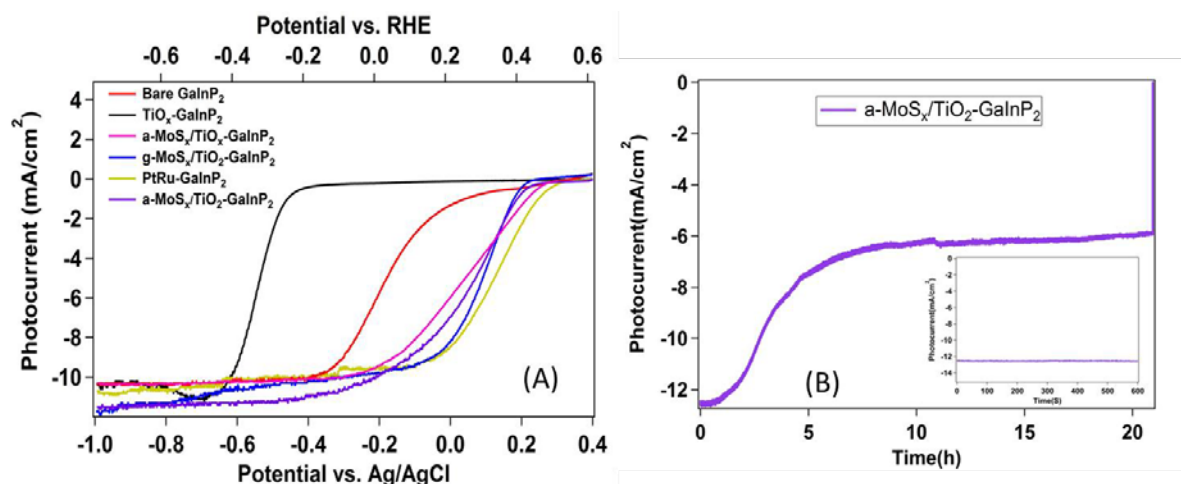
In the format provided by the authors and unedited.

## A graded catalytic-protective layer for an efficient and stable water-splitting photocathode

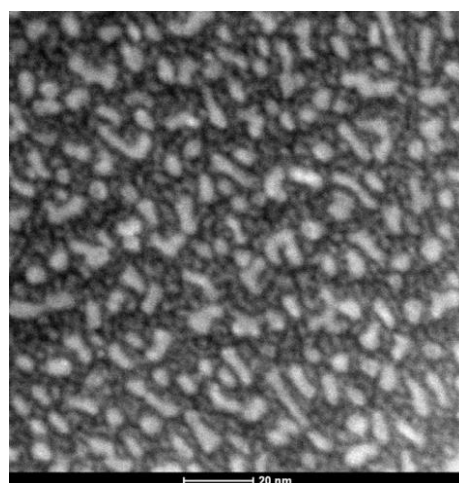
Jing Gu, Jeffery A. Aguiar, Suzanne Ferrere, K. Xerxes Steirer, Yong Yan, Chunxiao Xiao, James L. Young, Mowafak Al-Jassim, Nathan R. Neale, John A. Turner



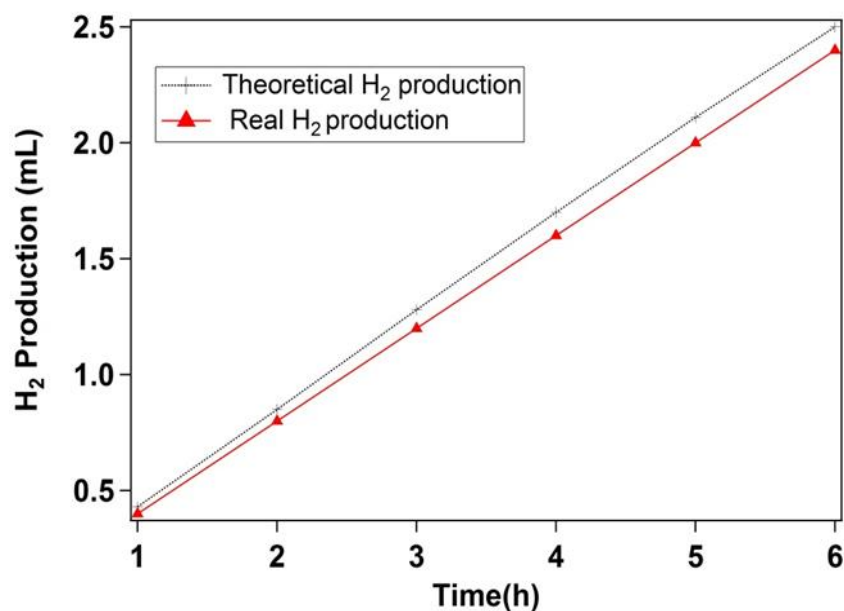
**Supplementary Figure 1.** (A) Current-Potential plots of unannealed a-MoS<sub>x</sub>-GaInP<sub>2</sub> compared with other electrodes; (B) 20 h photocathode stability of unannealed a-MoS<sub>x</sub>-GaInP<sub>2</sub> at 0 V vs. RHE (inset is 10 min electrolysis at 0V vs. RHE) (C) SEM image of a-MoS<sub>x</sub>-GaInP<sub>2</sub>



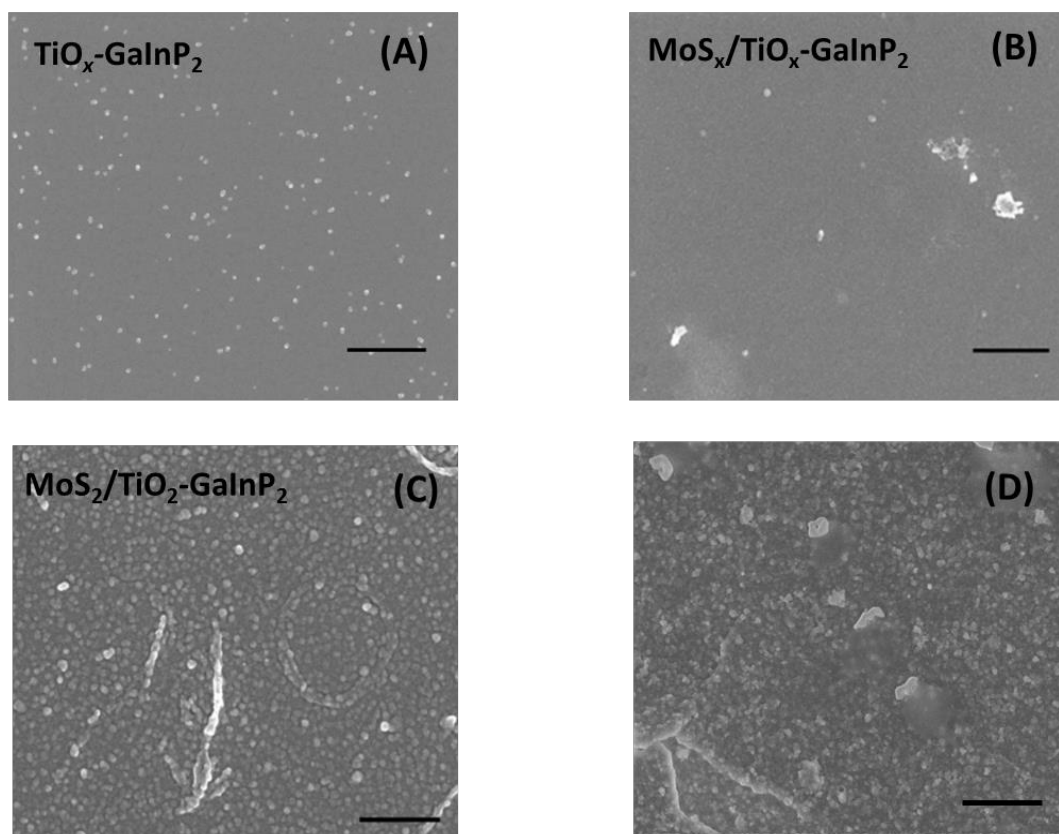
**Supplementary Figure 2.** (A) Current-Potential plots of a-MoS<sub>x</sub>/TiO<sub>2</sub>-GaInP<sub>2</sub> compared with other electrodes. (B) 20 h photocathode stability of a-MoS<sub>x</sub>/TiO<sub>2</sub>-GaInP<sub>2</sub> at 0 V vs. RHE (inset is 10 min electrolysis at 0 V vs. RHE)



**Supplementary Figure 3.** STEM HAADF image of PtRu-GaInP<sub>2</sub> electrode. Bright contrast regions are PtRu nanoparticles on the GaInP<sub>2</sub> surface. Under these imaging conditions, areas that have a higher average atomic number appear brighter.



**Supplementary Figure 4.** Calculated vs. experimental hydrogen production of g-MoS<sub>x</sub>/TiO<sub>2</sub>-GaInP<sub>2</sub> electrode under 1 sun illumination for the first 6 h electrolysis. The experiment was conducted in 0.5 M H<sub>2</sub>SO<sub>4</sub> under 1 sun illumination intensity at 0 V vs. RHE.



**Supplementary Figure 5.** SEM images of (A)  $\text{TiO}_x\text{-GaInP}_2$ ; (B)  $\text{a-MoS}_x/\text{TiO}_x\text{-GaInP}_2$ ; (C)  $\text{g-MoS}_x/\text{TiO}_2\text{-GaInP}_2$ ; and (D)  $\text{MoS}_2/\text{TiO}_2\text{-GaInP}_2$  after 20 h photoelectrolysis at 0 V vs. RHE (scale bar: 1  $\mu\text{m}$ ).

**Supplementary Table 1.** TON of TOF of a-MoS<sub>x</sub>/a-TiO<sub>x</sub>-GaInP<sub>2</sub> and c-MoS<sub>2</sub>/c-TiO<sub>2</sub>-GaInP<sub>2</sub> for 20 h electrolysis

| Sample                                                     | Catalyst amount (ppb) | Area of the electrode used (cm <sup>2</sup> ) | Catalyst loading (nmol/cm <sup>2</sup> ) | Total Charge in 20 h (Coulomb) | TON    | TOF (s <sup>-1</sup> ) |
|------------------------------------------------------------|-----------------------|-----------------------------------------------|------------------------------------------|--------------------------------|--------|------------------------|
| c-MoS <sub>2</sub> /c-TiO <sub>2</sub> -GaInP <sub>2</sub> | 161                   | 0.16                                          | 10.49                                    | 119                            | 3.67E5 | 5.1                    |
| PtRu-GaInP <sub>2</sub>                                    | 102                   | 0.15                                          | 3.5                                      | 74.25                          | 7.4E5  | 10.25                  |

**Supplementary Table 2.** Summary of activity and stability of photocathodes modified by molybdenum sulfide

| Reference                      | Light Absorber                                | HER catalyst                                   | $j_0$ (0 V vs. RHE, $\text{mA}\cdot\text{cm}^{-2}$ ) | Stability <sup>a</sup> at 0 V vs. RHE                        | Photo-current onset (vs. RHE) |
|--------------------------------|-----------------------------------------------|------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------|-------------------------------|
| Hu et al. <sup>1</sup>         | $\text{Cu}_2\text{O}/\text{AZO}/\text{TiO}_2$ | $\text{MoS}_{2+x}$                             | -5.6<br>-4.5<br>-2.0                                 | 5 h at pH 1<br>10 h at pH 4<br>10 h at pH 9                  | 0.45 V                        |
| Seeger et al. <sup>2</sup>     | Ti-n <sup>+</sup> p-Si                        | $\text{MoS}_x$                                 | -14                                                  | 1 h at pH 0<br>(+0.2 V vs. RHE)                              | 0.33 V                        |
| Tran et al. <sup>3</sup>       | Si-nano wires                                 | $\text{MoS}_2$                                 | -1                                                   | 1 h at pH 5.0                                                | 0.25 V                        |
| Hou et al. <sup>4</sup>        | Si pillars                                    | $\text{Mo}_3\text{S}_4$ cluster                | -9                                                   | 24 h<br>light/dark<br>cycle at pH 0                          | 0.15                          |
| Laursen et al. <sup>5</sup>    | n <sup>+</sup> p-Si                           | $\text{MoS}_2/\text{Mo}$                       | -10                                                  | 5 d in 1 M $\text{HClO}_4$                                   | 0.30 V<br>0.34 V              |
|                                |                                               | $\text{MoS}_x/\text{MoS}_2/\text{Mo}$          | -12                                                  | 24 h in 1 M $\text{HClO}_4$ (+0.2 V vs. RHE)                 |                               |
| Bourgeteau et al. <sup>6</sup> | P3HT: PCBM                                    | $\text{MoS}_x$                                 | -0.2                                                 | 45 min in 0.5 M $\text{H}_2\text{SO}_4$ (+0.16 V vs. RHE)    | 0.50 V                        |
| Zhou et al. <sup>7</sup>       | Si                                            | $\text{Pt}/\text{MoS}_2/\text{Al}_2\text{O}_3$ | -25                                                  | 40 h in 1 M $\text{HClO}_4$                                  | 0.30 V                        |
| Zhang et al. <sup>8</sup>      | Si                                            | $\text{MoS}_2/\text{TiO}_2$                    | -21                                                  | 75 min in 0.5 M $\text{H}_2\text{SO}_4$ (at -0.33 V vs. RHE) | 0.30 V                        |
| This work                      | $\text{GaInP}_2$                              | $\text{g-MoS}_x/\text{TiO}_2$                  | -11                                                  | 20 h at pH 0                                                 | 0.50 V                        |

a. Stability defined as after maximum electrolysis time, 70% or more of the initial photocurrent is maintained ( $j_t/j_0 \geq 70\%$ ).

**Supplementary References:**

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- 8 Zhang, L., Liu, C., Wong, A. B., Resasco, J. & Yang, P. MoS<sub>2</sub>-wrapped silicon nanowires for photoelectrochemical water reduction. *Nano Research* **8**, 281-287 (2015).